ProteinTuneRL: A Reinforcement Learning Perspective on Antibody Design

Algorithms, Correctness Analysis, and Training Behaviour

Ruman Shaikh

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Abstract

This report studies *ProteinTuneRL*, a framework that treats antibody CDR-loop design as constrained reinforcement learning (RL) over amino-acid tokens. An infilling protein language model (IgLM) provides the policy, and a cheap structural proxy provides the reward. Alongside the biological framing, the report makes three technical contributions. First, it adds and formalises an Advantage Actor-Critic algorithm with Generalized Advantage Estimation (A2C+GAE) that uses a token-level value function suited to variable-length infills. Second, it presents a correctness analysis of the online-RL machinery that identifies and repairs several defects: an incorrectly implemented adaptive KL controller, a support mismatch between the sampling distribution and the gradient estimate, a missing bootstrap for truncated episodes in GAE, an action-leaking value baseline in PPO, and an attention-kernel incompatibility that prevents REINFORCE from computing gradients. Each repair is covered by a regression test. Third, it examines a reward-design issue: the default β-sheet objective is a compositional amino-acid propensity rather than a structural prediction, and its signal is diluted when the score is computed over the entire chain. Training runs of PPO, A2C+GAE, and REINFORCE on a CDR-H3 β-sheet objective show that all three improve the reward, that they differ mainly in how far they drift from the pretrained distribution, and that most useful learning occurs within roughly one to two thousand steps. The central difficulty in this setting is controlling divergence from the pretrained biological prior while continuing to improve the design objective.

1 Introduction

Proteins are the functional units of biological systems. Their behaviour follows the principle that sequence determines a folded structure, which in turn determines function. Recent structure predictors have shown that machine learning can recover folds at scale, but accurate structure prediction alone does not solve protein *design*, which also depends on stability, binding behaviour, and evolutionary constraints.

Antibodies illustrate this difficulty. They bind foreign molecules through short, flexible *Complementarity Determining Regions* (CDRs), and binding is sensitive to small changes in sequence. Among the CDRs, CDR-H3 is the most variable and one of the most influential for binding, which makes it a natural target for sequence optimisation.

2 From Antibody Structure to Sequence Modelling

Antibodies have a characteristic Y-shape whose tips carry the CDR loops responsible for binding. Although a protein is a sequence, its function depends on folding, and computing the full three-dimensional fold of every candidate sequence is too expensive to place inside an optimisation loop. ProteinTuneRL therefore relies on proxy objectives, meaning cheap signals that correlate with structural behaviour. One such proxy is β-sheet propensity, an estimate of whether a sequence tends to form β-sheet-like structure. It does not guarantee correct folding, but it can be evaluated efficiently.

Because a protein can be written as a string over roughly 20 amino acids, it can be modelled with a language model. ProteinTuneRL is built around IgLM [1], an antibody language model designed for *infilling*: a region such as a CDR loop is masked, and replacements are generated conditioned on the surrounding context.

3 Antibody Design as Constrained Reinforcement Learning

Given a masked target region and a cheap proxy score, antibody infilling can be posed as a sequential decision problem over discrete symbols:

- **State**: the antibody sequence with a masked region and its surrounding context.
- **Action**: the choice of amino acid at each masked position.
- **Policy** π_θ: the language model's conditional distribution over amino acids.
- **Reward**: a scalar proxy (for example, β-sheet propensity) for the completed sequence.
- **Environment**: a deterministic evaluator that scores the completion.

Supervised learning is insufficient for this task. A model trained on existing antibodies reproduces the distribution of known sequences, which need not contain the optimal sequence for a specified objective. Reinforcement learning allows the policy to depart from imitation, while a **KL-divergence penalty** toward the pretrained IgLM [4] keeps generated sequences within a plausible region of sequence space. Writing π_{ref} for the frozen reference, each sampled sequence y receives

$$r'(y) \;=\; r(y) \;-\; \beta \left(\log \pi_\theta(y \mid x) - \log \pi_{\text{ref}}(y \mid x) \right), \tag{1}$$

so the objective balances exploration toward higher reward against retention of the pretrained distribution. The penalty is applied at the reward level and is therefore common to PPO, A2C, and REINFORCE.

4 What ProteinTuneRL Does in Practice

At a high level, ProteinTuneRL lets reinforcement learning operate on protein sequences without a physics-based simulation inside the training loop. The workflow is:

1. start from a real antibody sequence and keep most of it fixed;
2. mask a small region, usually a CDR loop;
3. use IgLM to fill in the missing amino acids;
4. score the generated sequence with a proxy objective;
5. apply reinforcement learning so later generations move toward higher scores while remaining close to the pretrained prior.

The system is driven by a single entry point that selects trainers, optimisers, and evaluators from configuration files. This modular structure makes it a suitable environment for comparing RL algorithms while the biological setup is held fixed.

5 Extensions Implemented

5.1 A2C + GAE

The main algorithmic extension is an Advantage Actor-Critic optimiser with Generalized Advantage Estimation [3]. The motivation is an alternative to PPO that is lighter to run, more stable than raw REINFORCE, and closer to classical actor-critic theory. The IgLM backbone acts as the actor. A copied backbone with a scalar linear head acts as the critic and produces a token-level value $V(s_t)$ for each infill position t; a single scalar value is inadequate for variable-length infills. The terminal reward r' is assigned to the last generated token, and GAE propagates it backward,

$$\delta_t = r_t + \gamma\, V(s_{t+1})\, m_{t+1} - V(s_t), \qquad A_t = \sum_{l \geq 0} (\gamma\lambda)^l\, \delta_{t+l}\,, \tag{2}$$

where m_{t+1} is a continuation mask, $\gamma = 0.99$, and $\lambda = 0.95$. The policy loss is $-A_t \log \pi_\theta$, the value target is $A_t + V(s_t)$, and an entropy term is added. Gradient norms of both actor and critic are clipped.

5.2 REINFORCE as a Baseline

The framework already includes REINFORCE, which is used here as a baseline. It applies the raw policy gradient with a mean-reward baseline and an entropy term, $\nabla_\theta J \approx \mathbb{E}[(r' - \bar{r})\, \nabla_\theta \log \pi_\theta(y \mid x)]$. The baseline makes the optimisation trade-offs of variance and stability easier to read in a relatively raw form.

5.3 Explicit KL-Regularised Optimisation

Maintaining proximity to the pretrained prior is a central concern in protein design, and the framework supports an explicit KL penalty between the updated policy and the original IgLM (Eq. 1). Increasing the KL weight constrains exploration and preserves biological realism; decreasing it allows more aggressive optimisation of the proxy reward. Because the penalty is applied at the reward level, it is compatible with PPO, A2C, and REINFORCE.

6 Correctness Analysis and Fixes

Extended training of the online algorithms exposed several defects in the shared machinery. Each was corrected and covered by a unit test (Table 1). The corrections apply to all three algorithms where the underlying code is shared.

Two of these defects determine whether an algorithm runs at all. Without gradient clipping, PPO can train for several thousand steps and then diverge to a non-finite state; clipping restores a stable trajectory. REINFORCE is incompatible with the eager attention kernel because it back-propagates through the autoregressive sampling graph, in which the attention mask is updated in place; changing the attention implementation removes the in-place operation. The remaining defects introduce bias rather than failure. The KL-controller and support-mismatch corrections make the KL regulariser quantitatively meaningful, and the prompt-only baseline restores an unbiased policy gradient.

Issue	Defect	Correction
Adaptive KL controller	$\texttt{KL} - \texttt{target}/\texttt{target}$ collapses to $\texttt{KL} - 1$ for any target; the update also made β a per-sequence tensor	Use $(\overline{\mathrm{KL}} - \tau)/\tau$ over the cross-process mean KL, and keep β a scalar synchronised across ranks
On-policy support	Sampling masks control tokens to $-\infty$, but the gradient recomputed log-probabilities over the full vocabulary	Apply the same constrained-support mask when recomputing log-probabilities and the KL estimate
GAE truncation	Episodes cut off by `max_length` were treated as terminal, so no value bootstrap was applied	Record a truncation flag and bootstrap $V(s_T)$ for truncated episodes only
PPO value baseline	The state-value baseline pooled over the whole sequence, including the generated action, which biases the gradient	Condition the value on the prompt tokens only
PPO stability	No gradient clipping was applied, and the policy could diverge to a non-finite state under sustained training	Add gradient-norm clipping and a non-degenerate mini-batch for advantage normalisation
REINFORCE backward	Under eager attention, an in-place update of the attention mask in the sampling loop breaks gradient computation	Use the scaled-dot-product attention implementation for the policy

Table 1: Defects in the online-RL machinery and their corrections. Pure-tensor unit tests cover the KL controller, the constraint masking, and the GAE bootstrap; end-to-end optimiser-step tests cover PPO and A2C.

7 Reward Design Considerations

The default objective is the β-sheet fraction returned by a standard biophysics library. This quantity is a compositional statistic. It counts the fraction of residues whose amino-acid identity belongs to a fixed sheet-favouring set, and it is not a prediction of secondary structure. Optimising it rewards the emission of particular residues, which correlates only loosely with the folded structure and is a weak proxy for a CDR-H3 loop that does not adopt a β-sheet conformation.

A second issue is dilution. The score was computed over the concatenated heavy and light chains, about 230 residues, whereas the policy controls only the roughly 12 infilled CDR-H3 positions. The learnable signal is therefore a small fraction of a metric dominated by the fixed framework. We add a `region_only` option that scores only the infilled span, which removes the dilution. The normalisation constants must then be recomputed for the region-only distribution, and its larger magnitude makes gradient clipping advisable in sensitive optimisers such as PPO. Structural rewards based on predicted folding remain available at a substantially higher cost per step.

8 Results and Training Behaviour

This section characterises the behaviour of the online algorithms rather than reporting best-case performance. Each run optimises the CDR-H3 β-sheet objective for 2000 steps at batch size 8 under the KL regulariser of Eq. 1. Two quantities are tracked: the training reward, which already includes the negative KL term, and the KL divergence from the pretrained IgLM, which measures policy drift. All curves use a 200-step moving average.

8.1 PPO

PPO (Figure 1) behaves conservatively. The reward rises early and then settles onto a stable plateau, and the KL increases and then stabilises rather than growing without bound. Training is smooth, with no large oscillations. This is the expected behaviour of a clipped update that trades update size for stability in a high-dimensional discrete action space.

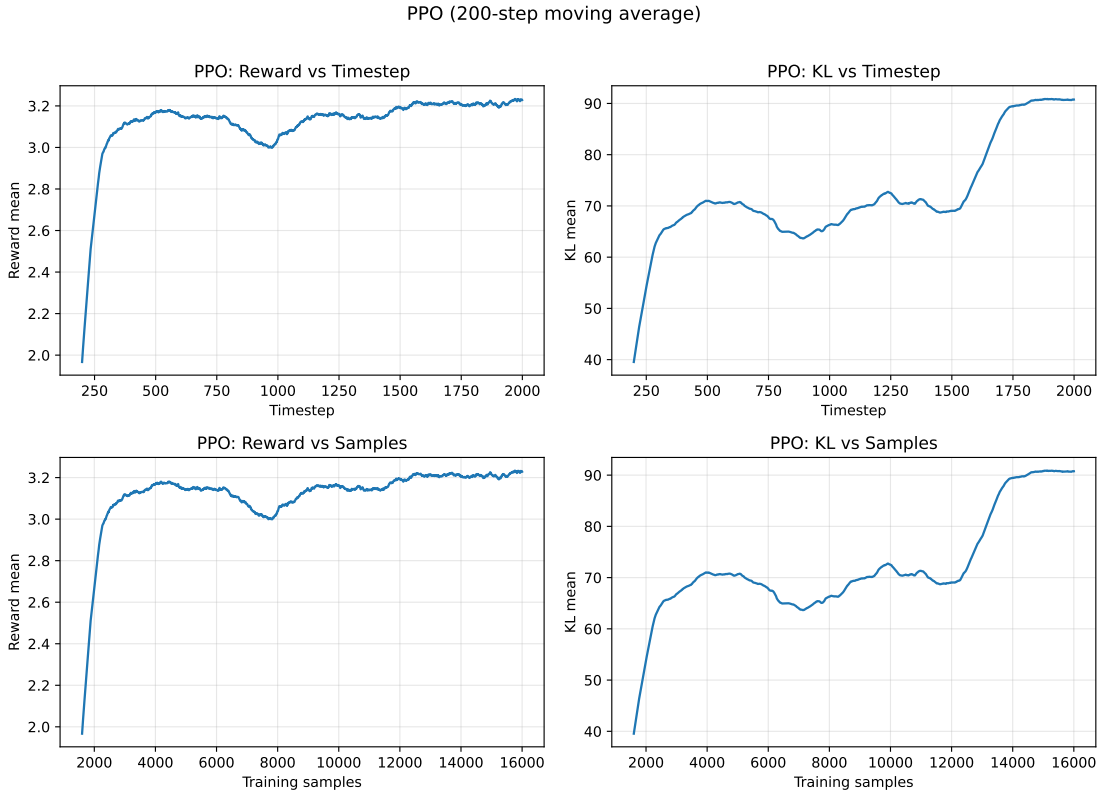


Figure 1: PPO: reward and KL divergence against timestep and against training samples (200-step moving average). Reward plateaus while KL stabilises.

8.2 A2C + GAE

A2C with GAE (Figure 2) learns with sharper dynamics. The reward climbs quickly early in training, consistent with the improved credit assignment of the token-level critic, and the KL grows more aggressively than for PPO. In these runs A2C reaches a comparable reward at roughly half the final KL divergence of PPO and REINFORCE, which indicates that the advantage estimate

concentrates updates where they are informative. Without PPO-style clipping the policy retains more freedom to drift later in training, which is the reason gradient clipping is applied to the actor and critic.

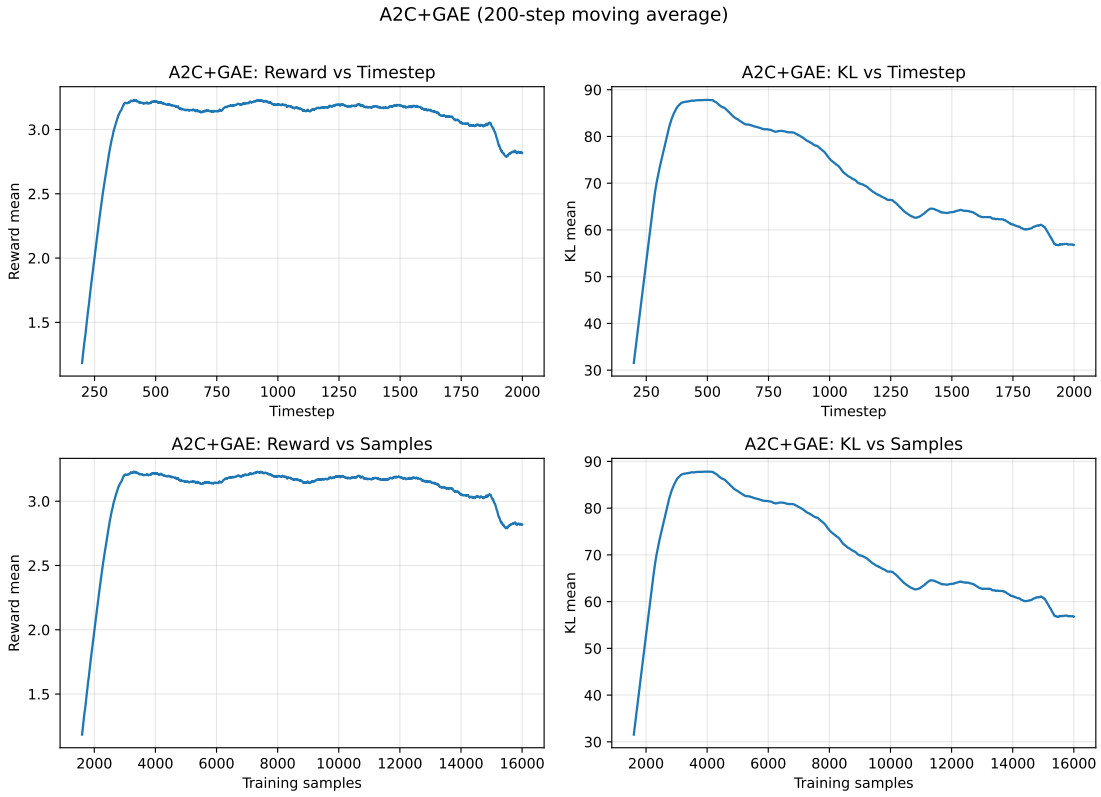


Figure 2: A2C+GAE: faster early reward gains and a lower final KL than PPO and REINFORCE in these runs, with a greater tendency to drift late in training.

8.3 REINFORCE

REINFORCE (Figure 3) behaves as a high-variance policy-gradient baseline. The reward improves but noisily, the KL grows steadily with limited self-correction, and convergence is slower and less reliable than for PPO or A2C. This baseline confirms that the optimisation signal is present and shows why variance reduction and trust-region constraints are useful in sequence design.

8.4 Explicit KL-Regularised REINFORCE

Increasing the KL weight changes REINFORCE substantially (Figure 4). The KL is actively controlled and settles at a lower level than the unregularised baseline (about 56 against 84 in these runs), the reward plateaus earlier and at a slightly lower value, and training becomes more stable. This makes the underlying trade-off explicit: optimisation now balances reward against proximity to the biological prior.

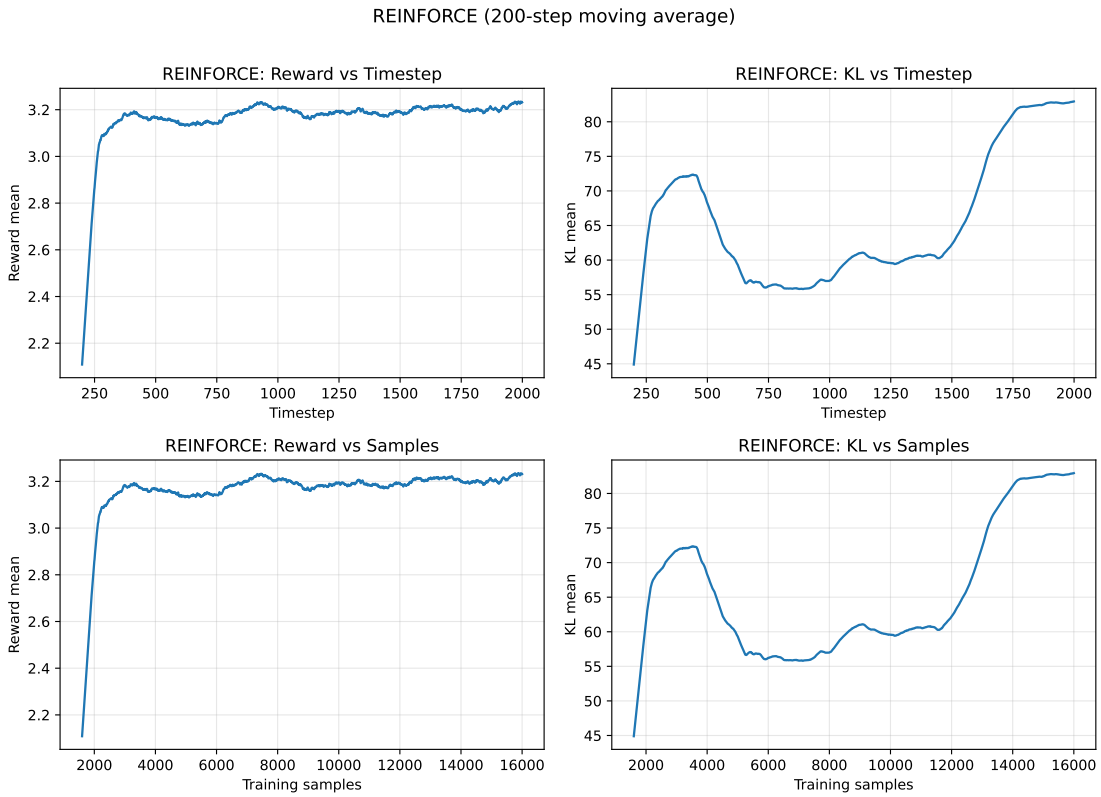


Figure 3: REINFORCE: noisy reward improvement and steadily growing KL, reflecting the high variance of the raw policy gradient.

8.5 Convergence and Step Budgets

Across the online algorithms, most of the reward improvement occurs within roughly the first one to two thousand steps. Beyond that point the reward gains saturate and the KL dynamics increasingly dominate, so further training mostly trades biological realism for marginal proxy improvement. For this task and this proxy, early optimisation captures most of the useful signal.

9 Conclusion

ProteinTuneRL treats antibody design as constrained exploration in sequence space. This report contributes a precise token-level A2C+GAE formulation for variable-length infills, together with a correctness pass that makes the KL regulariser, the value baseline, and the credit-assignment machinery behave as specified. Several previously silent biases and one gradient-computation failure are now covered by tests. The reward analysis shows that a tractable proxy must be interpreted with care, because optimising a compositional statistic does not guarantee an improvement in the folded structure. Across PPO, A2C, and REINFORCE the algorithms differ mainly in how far they drift from the pretrained distribution, which supports the view that the central difficulty here is controlling divergence from the biological prior while making measurable progress on the design objective.

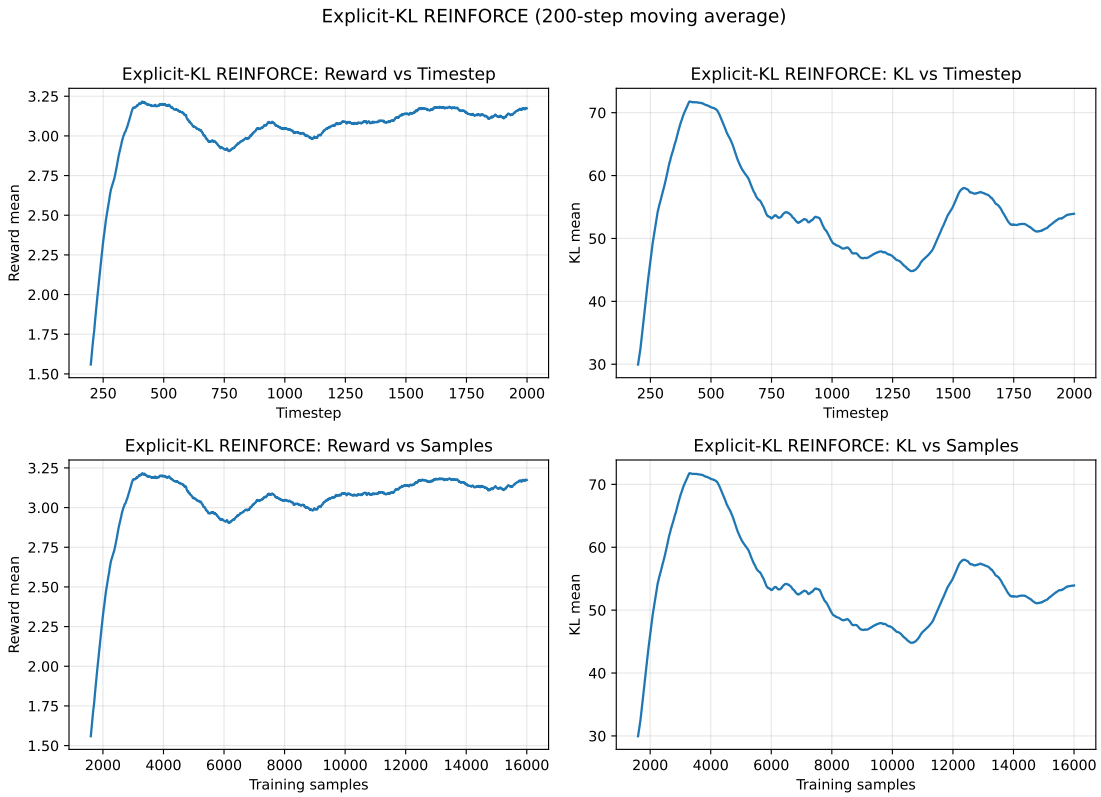


Figure 4: Explicit-KL REINFORCE: a stronger KL penalty holds the divergence lower and stabilises training, at the cost of a slightly lower reward plateau.

10 Future Work

Several directions follow from this study. A more faithful reward would score predicted structure rather than amino-acid composition; folding-based confidence is already available in the framework but is far more expensive per step, so efficient approximations are worth exploring. The region-only reward should be recalibrated so that its normalisation matches the infilled-span distribution. Multi-objective and constrained formulations would let developability and binding proxies be optimised jointly rather than through a single scalar. Critic-free, group-relative estimators offer a lighter alternative to the token-level critic for very short episodes. Finally, the proxy objectives should be validated against measured biological properties on a held-out set, so that proxy improvement can be linked to real design quality.

Acknowledgements

This project builds on the ProteinTuneRL framework and on IgLM as the underlying antibody language model. The algorithmic extensions, correctness analysis, and experiments reported here were carried out on top of that codebase.

References

[1] R. Shuai, J. Ruffolo, J. Gray. *Generative language modeling for antibody design.* Cell Systems, 2023.

[2] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov. *Proximal Policy Optimization Algorithms.* arXiv:1707.06347, 2017.

[3] J. Schulman, P. Moritz, S. Levine, M. Jordan, P. Abbeel. *High-Dimensional Continuous Control Using Generalized Advantage Estimation.* arXiv:1506.02438, 2015.

[4] D. M. Ziegler et al. *Fine-Tuning Language Models from Human Preferences.* arXiv:1909.08593, 2019.